

Import Data using Transform Maps (Spreadsheet)

ServiceNow Project Documentation

Phase 3: Project Design Phase

Item	Details
Project Title	Import Data using Transform Maps (Spreadsheet)
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Platform	ServiceNow
Team ID	SWTID-2026-2373

3.1 Problem–Solution Fit

Problem Statement

Organizations often receive bulk employee data in Excel or spreadsheet format from HR teams or external systems. Entering this data manually into ServiceNow causes several problems:

- It is slow and needs a lot of manual effort for large volumes of data.
- It is prone to typing errors, wrong values and inconsistent formats.
- Repeated imports of the same data create duplicate records.
- There is no easy way to update existing records when employee details change (for example, a name or email change).
- Administrators and HR teams have no quick visual summary of employee data.

Solution Fit

Problem	Solution in This Project
Manual, slow data entry	Bulk upload of the Excel file using Load Data into an Import Set staging table
Source and target fields don't match	Transform Map with field mapping (Auto Map Matching Fields and Mapping Assist)
Duplicate records on repeated imports	Coalesce on Employee ID, so existing records are updated and duplicates are ignored
Data changes over time	Re-importing updated sheets updates records (inserts: 2, updates: 2)
No visibility of employee data	Reports (Pie, Bar, List) combined into the Employee Analytics Dashboard

3.2 Proposed Solution

Aspect	Description
Approach	A staged import: Spreadsheet → Import Set table → Transform Map → Target table
Source data	Sample Spreadsheet.xlsx with 5 columns: Employee ID, Name, Email, Department, Location (15 initial records)
Staging table	Employee Import (u_employee_import), created through Load Data
Target table	Employee Test (u_employee_test), a custom table with 5 String fields
Transformation	Transform Map "Sample Spreadsheet Import" mapping the 5 source fields to the 5 target fields
Duplicate control	Coalesce enabled on u_employee_id
Validation	Transform History (Total, Inserts, Updates, Ignored, Skipped, Errors) and the Employee Tests list view
Insights	3 reports (Employees by Department, Employees by Location, Employee List Report) added to the Employee Analytics Dashboard

Key Features

1. Bulk import of Excel data with no manual entry.
2. Field-level mapping between the staging and target tables.
3. Coalesce-based insert-or-update logic, which keeps data clean.
4. Import and transform results verified through the progress and history screens.
5. Visual reporting and a shareable dashboard, which can be restricted by users, groups and roles.

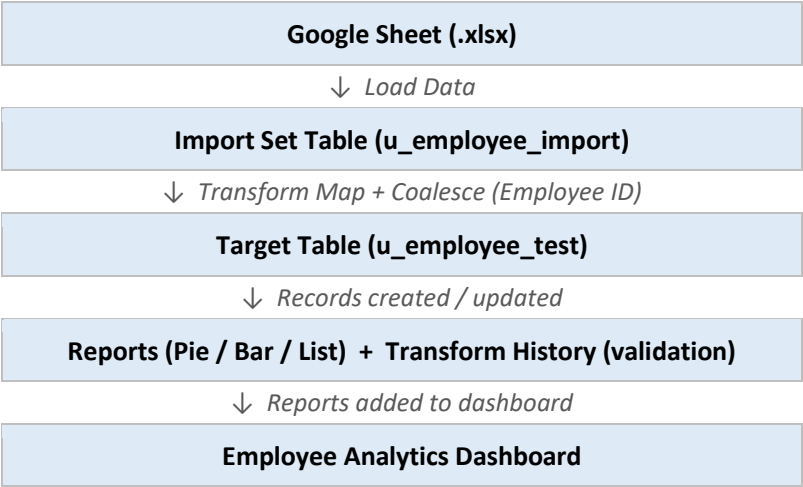
3.3 Solution Architecture

Architecture Layers

Layer	Component	Purpose
Data source	Google Sheets / Excel (.xlsx)	Holds the raw employee data
Ingestion	System Import Sets → Load Data	Uploads the file into a staging table
Staging	Import Set table u_employee_import	Temporary holding area for the raw data
Transformation	Transform Map "Sample Spreadsheet Import"	Maps fields, runs the transform, applies coalesce
Storage	Custom table u_employee_test	Final store of validated employee records
Validation	Import log, Transform History, list view	Confirms inserts, updates and errors

Layer	Component	Purpose
Presentation	Reports → Dashboard (PA_DASHBOARDS)	Charts and lists for HR and admins

Data Flow



Field Mapping Design

Source field (Import Set)	Target field (Employee Test)	Coalesce
u_employee_id	u_employee_id	true
u_name	u_employee_name	false
u_email	u_email	false
u_department	u_department	false
u_location	u_location	false

Import Behaviour Design (Coalesce Scenarios)

Scenario	Total	Inserts	Updates	Ignored
First import (15 records)	15	15	0	0
Second import (2 new + 2 changed)	4	2	2	0
Same sheet re-imported	4	0	0	4

Reporting Design

Report	Type	Group by / Columns	Aggregation
Employees by Department	Pie	Department	Count
Employees by Location	Bar	Location	Count
Employee List Report	List	Employee ID, Name, Email, Department, Location	N/A

Design Considerations

- Security: The pa_dashboards create ACL needed Admin Overrides set to true to open the dashboard form by its backend name (PA_DASHBOARDS.FORM). Dashboard sharing can be limited by users, groups and roles.
- Data integrity: A staging table plus coalesce avoids direct writes of unvalidated data to the target table.
- Scalability: The same Transform Map can be reused for later loads into the same Import Set table.